

EMG Analysis for Wrist-Hand Motion Classification

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Abstract

This dissertation investigates the use of surface electromyography (EMG) signals for two key applications in Human-Computer Interaction (HCI): hand gesture classification and biometric user identification. Using the publicly available UCI EMG dataset, two machine learning pipelines were developed and evaluated.

The first pipeline restructured an existing approach to classify wrist-hand gestures, introducing improved preprocessing, feature extraction, and class balancing methods. The resulting model achieved a classification accuracy of approximately 99%, outperforming the original implementation. Additional experiments examined the impact of varying EMG channels and segmentation window sizes on performance across two different models.

The second pipeline, designed for biometric user identification, employed time-domain features and Kernel Fisher Discriminant Analysis for dimensionality reduction. A multi-layer perceptron (MLP) classifier achieved a peak accuracy of 96.46% across 15 users, exceeding prior benchmarks. Analysis also indicated that specific gestures yielded more distinctive biometric signatures.

The results highlight the potential of EMG signals for gesture recognition and biometric authentication in HCI.

Specification

1. Setup a machine learning pipeline for EMG analysis in Python
2. Explore the variation in performance of the pipeline with features.
3. Explore the variation in performance with numbers of EMG channels, and EMG features.
4. Define a low-cost EMG biometric regime which maintains performance but with reduced complexity.

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My heartfelt gratitude to my parents for the love and encouragement towards the completion of this work.

Declaration of Originality

I understand the nature of plagiarism, and I am aware of the University's policy on this.

I declare that this report is my original work except where stated.

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1. Introduction

1.1 Background and Motivation

The combination of human-computer interaction (HCI) and biometric systems offers a promising opportunity for developing secure and intuitive interfaces. Among the many HCI approaches, Hand Gesture Recognition (HGR), stands out as an expressive and natural method, it enables machines to interpret hand movements and respond accordingly by predicting the exact movement the hand is performing.

Electromyography (EMG) signals generated by the electrical activity of muscles provide a rich source of information that can be utilised to make predictions and interpret wrist-hand movement, they also carry unique signatures of individual muscular activity which can be utilised for biometric user identification[1], [2]. With these qualities, EMG is an invaluable tool for both movement interference and user identification in biometric systems. However, these tasks are challenging due to the complexity of EMG signal processing, individual variability, and the need for efficient feature extraction and machine learning techniques to achieve reliable performance.

There are two approaches for acquiring EMG data, the first approach is to place EMG sensors on specific known places on the arm muscles. This approach is widely used in clinical and research settings since it offers high signal quality and minimal interference from surrounding muscles. The targeted placement of sensors benefits the accuracy of detection which also enables the use of sophisticated signal processing techniques. This makes it ideal for high precision applications but also requires knowledge of anatomy and manual setup, limiting its practicality in a real-world everyday system.

The other approach is to place EMG sensors randomly on the arm[1], [3]. EMG devices like the MYO armband enables us to use wearable sensors that simplify the acquisition of EMG signals[4], [5]. These wearable devices do not require precise muscle targeting, making them ideal for real-time HCI applications and biometric systems. However, they may struggle from interference from adjacent muscles affecting signal fidelity. This makes data preprocessing and feature extraction critical for robust performance.

For this project, the UCI data set was used as the main source of data. It is a time-series data collected via a MYO armband; the data consists of raw EMG recordings for 36 subjects while they perform a series of hand gestures[6]. The focus is on working within the limitations of the random placement approach, considering realistic constraints like signal variability and limited precision. The aim is to build robust, accurate and low-complexity models that could function in practical wearable systems without the need for expert calibration.

1.2 Problem Statement

Designing reliable EMG based system can be faced with challenges related to signal noise, user variability and system complexity. Feature extraction from raw EMG data critically affects model performance, varying different number of features and channels has an impact on the performance and efficiency of the model[7], [8].

This project aims to develop an EMG-based system that balances low computational complexity with high predictive performance, this can be achieved by evaluating the contribution of features, evaluating contributions of channels and measuring performance vs. complexity[9]. This dual focus on functionality and efficiency is relevant for applications that require lightweight, real-time EMG systems, such as immersive environments, video games and bionic limbs[10].

In addition to gesture recognition, the project addresses biometric user identification using EMG data. This will involve the implementation of a new machine learning pipeline inspired by research that shows the potential of time-domain pattern recognition for accurate biometric identification[11].

Due to the relatively limited length and variability of the UCI dataset, the project also considers data scarcity as a core constraint, evaluating how well the model generalise with limited or gesture-specific training conditions. A gesture specific approach is proposed to identify the most discriminative movements for user identification.

1.3 Aims and Objectives

This project aims to:

- Build a robust machine learning pipeline for wrist-hand gesture recognition using EMG signals.
- Design and implement a novel user identification pipeline based on EMG biometrics, using time-domain pattern recognition.
- Systematically evaluate performance across different:
 - Feature subset and EMG channel combinations.
 - Gestures and gesture sets.
 - Number of users and gesture-specific biometric scenarios.
- Explore practical deployment scenarios with re-identification periods (e.g., every 5 seconds) and the limitations posed by realistic movement patterns and data availability.
- Achieve state of the art accuracy with the lowest EMG complexity of its kind to enable real-time applications.

1.4 Contribution of This Work

This project presents two machine learning pipelines for Electromyography (EMG)-based human-computer interaction: one for wrist-hand gesture recognition and another for biometric user identification, both trained on the same UCI dataset and built for deployment on wearable systems using randomly placed EMG sensors.

The gesture recognition pipeline was redesigned from a legacy implementation, implementing advanced data balancing techniques (such as BorderlineSMOTE and Edited Nearest Neighbours), improved preprocessing, and model optimisation by conducting hyperparameter tuning. These enhancements increased classification accuracy from 95% in the original implementation to ~ 99%, while also maintaining a low model complexity.

A user identification pipeline was developed from scratch, inspired by research on time-domain EMG biometrics. The system achieved higher identification accuracy than the results published in the reference study[11], this was done by experimenting with different preprocessing techniques and gesture specific training. This validates the effectiveness of proposed feature extraction and classification strategy.

1.5 Overview of Dissertation

This dissertation is organised into six chapters, each addressing a core component of the research on EMG-based hand gesture recognition and biometric user identification

Chapter 1: Introduction outlines the research motivation, objectives, and contributions, framing the dual focus on gesture classification and biometric authentication using surface EMG signals.

Chapter 2: Literature Review examines prior work in EMG-based gesture recognition and biometric systems.

Chapter 3: Dataset describes the UCI EMG dataset used throughout the project, detailing its structure, signal properties, gesture classes, and limitations such as class imbalance.

Chapter 4: Gesture Classification Pipeline presents the redesigned classification pipeline, covering preprocessing, feature extraction, model training, and evaluation. It also explores how model performance varies with different EMG channel subsets and window sizes.

Chapter 5: User Identification Pipeline introduces a novel biometric identification system using forearm EMG signals. It discusses feature selection, dimensionality reduction via Kernel Fisher Discriminant Analysis, classification using a multi-layer perceptron, and results including gesture-specific performance insights.

Chapter 6: Conclusion and Future Work summarises the main findings, acknowledges system limitations, and outlines future directions, including real-time implementation, and the integration of both pipelines into a unified system.

2. Literature Review

2.1 Hand Gesture Recognition Techniques

Human-Computer Interaction (HCI) has evolved towards more seamless and intuitive communication between humans and machines. Hand gesture recognition stands out among these as a natural, expressive, and not intrusive technique. It enables systems to interpret the physical gestures of a user's hand or wrist and translate them into actionable commands, this makes it ideal for many applications such as immersive environments, assisted technology, and wearable devices.

There are three primary methods used for gesture recognition: vision-based system, inertial sensing (IMUs), and electromyography (EMG)-based systems.

Vision based systems relies on cameras to visually track and interpret hand movements in 2D or 3D space. These systems are commonly found in gaming, AR/VR, and robotics applications, they can capture both static poses and dynamic sequences. However, they often suffer from practical limitations in real-world settings, such as sensitivity to lighting conditions, occlusions, and background noise [12], [13]. While vision-based methods can provide detailed motion analysis, they may not be robust enough for all-day wearable or more applications.

Inertial measurement units can be used in systems. IMU-based approaches involve accelerometers and gyroscopes to measure movement and orientation. These are lightweight and power-efficient which makes them suitable for portable or embedded systems. However, they face limitations as drift, susceptibility to noise, and the need for calibration. These limitations impact accuracy and require careful consideration to their implementation[14].

EMG-based HGR, the focus of this project, uses electrical activity generated by muscle contraction to infer user intent. Unlike vision or IMUs, EMG can detect the user's motion before it becomes physically observable, this makes it a powerful tool for real-time systems that need anticipatory control and other applications[15], [16].

EMG signals can be used for wearable HCI applications and biometric systems, this is due to the access to fine motor control, which allows for subtle hand movements to be captured. It also contains unique muscle activation patterns that vary across individuals, making it suitable for user identification. However, EMG systems have challenges. Signals are often noisy and vary significantly between individuals and sessions. To address this, the configuration of sensors becomes a key design decision.

There are two main strategies for sensor placement in EMG based systems:

Targeted placement, which involves positioning electrode over specific forearm muscles such as the flexor carpi ulnaris or extensor digitorum. This method provides high quality, clean EMG signals and allows for sophisticated processing techniques.

However, it requires anatomical knowledge and manual setup, which can reduce usability in practical, everyday systems[17].

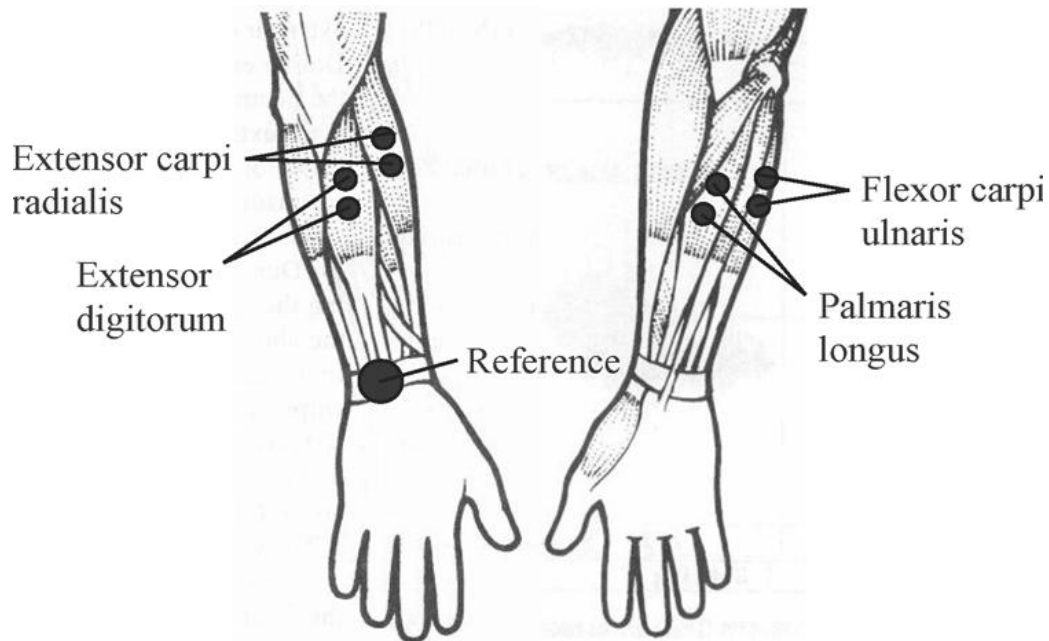


Figure 1 EMG placed sensors over specific arm muscles

Fixed/random placement, uses a fixed array of sensors placed uniformly around the arm. Devices like the MYO armband adopt this approach, using eight evenly spaced sensors to capture signals without needing expert calibration. While more convenient for users, this method introduces crosstalk between muscles and generally lower signal-to-noise ratio, making preprocessing and feature extraction critical for accuracy. Fig [x] shows the randomly placed armband over the forearm skin's surface and the interior cross-section view of upper forearm muscles.

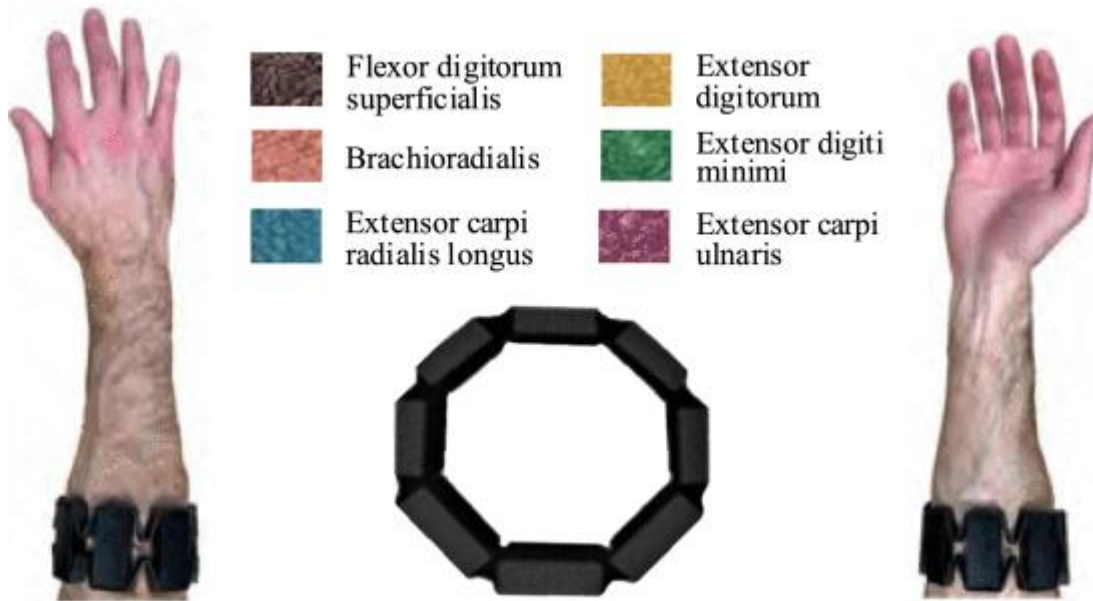


Figure 2 Randomly placed armband over the forearm skin's surface

2.2 Electromyography in Human-Computer Interaction and Biometric Systems

Electromyography (EMG) signals have emerged as a compelling input modality for Human-Computer Interaction and biometric systems. They have the capacity to capture muscular activity directly from the user, making them feasible for scenarios where traditional interaction methods are limiting, such as immersive environment, video games, electronic gadgets or control of robotic devices or 'bionic' limbs[18].

In HCI, EMG provides direct mapping of user intent to system actions. EMG signals offer anticipatory control since they're generated during muscle activation phase that precedes visible movement, unlike vision or IMU-based systems that rely on observable motion. This enables a more responsive and fluid interaction, especially in applications requiring real-time feedback such as virtual reality control, exoskeletons, or gesture based operating systems[19].

In biometric applications, EMG signals possess distinct individual specific muscle activation patterns that are unique and difficult to replicate. This makes EMG-based biometrics a wide area for future research and applications. The uniqueness of the EMG signals arises from the psychological and neurological differences such as in muscle structure, motor unit recruitment, and signal propagation[20].

EMG's non-invasive nature aligns with the need of wearable technology. Devices like the MYO armband demonstrate how EMG can be embedded into consumer-friendly

form factors, enabling passive, continuous authentication without requiring active user input[11], [21]. However, the integration of EMG in practical systems presents challenges. Variability across users and sessions, and signal noise due to movement and respiration introduce uncertainty. This highlights the importance of robust preprocessing, feature extraction, adaptive algorithms and evaluation strategies, issues addressed throughout this work.

2.3 Machine Learning Approaches for EMG-Based Classification

EMG signals are electrical signals generated by skeletal muscles in response to stimulation from motor neurons, which originates in the central nervous system (CNS). These myoelectric signals reflect neuromuscular activity and have applications ranging from clinical diagnostics to advanced control systems. In human-computer interaction, EMG signals are useful due to their ability to capture subtle muscle movements and temporal patterns, which can be used in gait analysis, prosthetic control, and gesture recognition[22].

A typical EMG-based classification pipeline begins with preprocessing the data, which addresses the noise, drift, and baseline variations. This is followed by windowing, where the continuous EMG stream is segmented into overlapping time windows for localised analysis. From these segments, features are extracted, either handcrafted (in the time or frequency domain) or automatically using deep learning techniques, and passed to classifier to make the predictions.

Several types of classifiers have been successfully applied to EMG signals.

Traditional machine learning algorithms such as Support Vector Machine (SVM), k-Nearest Neighbours (k-NN), Random Forests (RF), and Multi-Layer Perceptrons (MLP) have shown strong performance with limited data, making them well-suited for real-time and resource constrained applications[23]. These models are relatively low in complexity and require minimal training data.

Some approaches employ the use of deep learning models such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. These methods are proficient at capturing temporal dependencies and spatial correlations in the signal, resulting in improved accuracy. However, they often require more data and computational resources[24].

2.3.1 EMG-Based Hand Gesture Recognition (HGR)

Hand Gesture Recognition (HGR) refers to the process of identifying hand or wrist movements from a predefined set of gestures using muscle signals. A commonly adopted approach involves the use of surface electromyography (sEMG), which records the electrical activity generated by muscles through non-invasive sensors placed on the skin.

A typical EMG-based HGR system follows a structured pipeline involving several stages. The first step is data acquisition, where signals are captured using wearable sensor arrays, such as armbands, which measure voltage fluctuations resulting from muscle activity. These signals are sampled at high frequencies, generally ranging between 200 and 1000 Hz, to accurately capture the fast dynamics of muscle contractions.

Following acquisition, the raw EMG signals undergo pre-processing. Since these signals are inherently noisy, standard cleaning techniques such as bandpass filtering (usually between 20–450 Hz), rectification through taking the absolute value, and normalisation are applied to enhance signal quality and consistency across samples.

The cleaned signal is then segmented into short overlapping time windows, typically between 150 and 300 milliseconds in duration. Each window is treated as an independent instance for classification, allowing the model to capture temporal features of muscle activity without being overwhelmed by the full continuous stream.

After segmentation, each window is subjected to feature extraction to convert the raw data into a compact, informative representation. Time-domain features such as Mean Absolute Value (MAV), Root Mean Square (RMS), Variance, Waveform Length, Zero Crossings, and Slope Sign Changes are commonly used. Frequency-domain features, including Median Frequency, Mean Frequency, and Spectral Entropy, are often derived using techniques like the Fast Fourier Transform (FFT). In addition, time-frequency features can be extracted using methods like the Short-Time Fourier Transform (STFT) or Wavelet Transform to capture both the temporal and spectral characteristics of the signal[25].

The extracted features are then fed into classification models that associate them with specific gesture labels. Traditional machine learning approaches include k-NN,

SVM, Random Forests, and Linear Discriminant Analysis (LDA). More recent systems use neural networks such as Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNNs) for recognising spatial patterns, and Long Short-Term Memory (LSTM) networks for modelling temporal sequences[23][24].

To enhance robustness, post-processing techniques may be employed. These can include majority voting, temporal smoothing, or confidence-based rejection strategies to refine the final predictions and reduce misclassification.

Depending on the intended use, classification models may be designed to be user-specific, which means they are trained on data from a single individual and offer high accuracy at the cost of requiring personal calibration. Alternatively, models can be user-independent, trained on data from multiple users to provide broader generalisation, though often with a slight performance trade-off due to inter-user variability.

Latency is an important consideration in real-time applications. For effective responsiveness, gesture prediction must typically occur within 100 to 300 milliseconds after gesture initiation. Factors contributing to this latency include the window size used for segmentation, the time required for feature computation, and the prediction speed of the model.

2.3.2 EMG-Based User Identification

EMG signals possess unique individual-specific patterns, making them a promising candidate for biometric user identification. This application is relevant in secure access control, human authentication, and personalised of wearable systems.

Unlike traditional psychological signals such as ECG or EEG, surface EMG allows users to actively engage with the system using customisable hand gestures, which can enhance identification accuracy and user control. Time-domain and frequency-domain features are extracted from multi-channel EMG data to capture the temporal dynamics and spectral characteristics of muscle activity. In [26], a wearable multi-channel EMG armband (MYO armband) was used to extract feature across both domains and evaluate the identification performance using classifiers such as k-NN, LDA, and ensemble models. The best result, achieved by using k-NN yielded an accuracy of 86.01%.

Other experiments have shown that identification accuracy decreases with larger user groups, with accuracies ranging from 90.32% for 4 users to 55.32% for 19 users, indicating the impact of inter-user variability[27]. Gesture selection also plays a significant role, as some gestures lead to higher classification performance than others. Additionally, short term windows (e.g., 0.8 seconds) have shown to be sufficient for accurate identification, making sEMG viable for real-time applications[28]. In this work, we will evaluate how identification accuracy is impacted by different number of users and different gestures preformed.

3. Dataset

3.1 UCI EMG Dataset

The UCI EMG dataset forms the foundation of the gesture classification and user identification work as both pipelines were trained on the UCI dataset[6]. This section describes the data collection methodology, participant characteristics, and signal properties.

The dataset contains recordings from 36 participants performing hand gestures while wearing a MYO Thalmic bracelet on their forearm. Each participant completed two recording sessions, with data collected from both arms (indicated by column indices 1 and 2). This dual-arm approach provides valuable insights into the differences between dominant and non-dominant limbs and supports the development of arm-independent classification systems.

Participants performed a series of static hand gestures, each maintained for 3 seconds followed by a 3-second rest period. This protocol ensured consistent data collection across all subjects. The labels include:

0. Unmarked data
1. Hand at rest
2. Hand clenched in a fist
3. Wrist flexion
4. Wrist extension
5. Radial deviation
6. Ulnar deviation
7. Extended palm (not performed by all subjects)

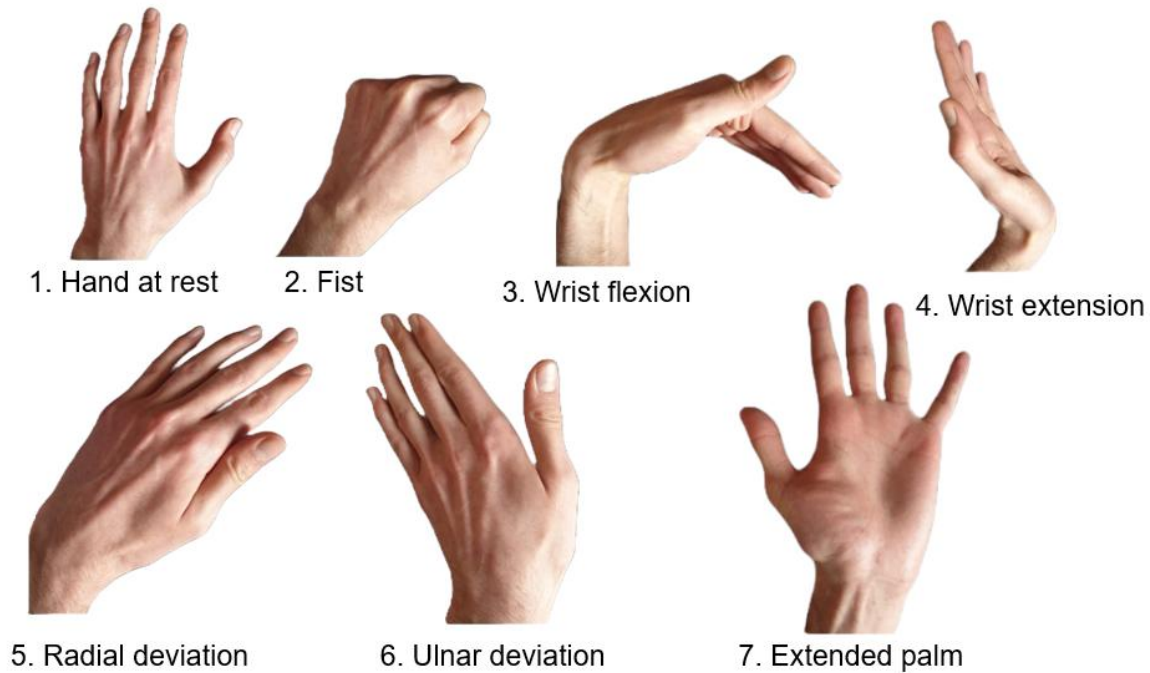


Figure 3 Gestures performed in UCI dataset

Unmarked data (class 0) was recorded during transitions between gestures.

3.1.1 EMG Channels

The MYO Thalmic bracelet used for data collection has eight EMG channels that are positioned circumferentially around the subject's forearm. These eight sensors simultaneously record myographic activity, providing comprehensive coverage of muscle activation patterns during gesture execution.

The electrode configuration ensures that signals are captured from various muscle groups in the forearm, including both flexor and extensor muscle groups, which are primarily responsible for finger and wrist movements. This comprehensive coverage is essential for distinguishing between similar gestures that may activate different combinations of muscles.

3.1.2 Data Structure

Each data file in the UCI EMG dataset is organised in tabular format and contains eleven columns representing temporal, physiological, and contextual information. These columns are as follows:

- Time: A timestamp recorded in milliseconds, indicating the precise time at which each EMG sample was captured.

- Channels 1–8: Eight columns corresponding to the raw EMG signals acquired from the eight sensor channels of the MYO armband.
- id: A numerical identifier representing the subject from whom the data was collected. Each subject has a unique ID.
- idx: The recording session identifier, which indicates whether the data was collected from the subject's left or right arm. This value is either 1 or 2.
- Class: The label assigned to the recorded gesture.

The EMG signals have been normalised, ensuring consistent amplitude ranges across recordings. This normalisation reduces the impact of individual differences in signal strength and facilitates more reliable feature extraction and classification.

3.1.3 Class Distribution

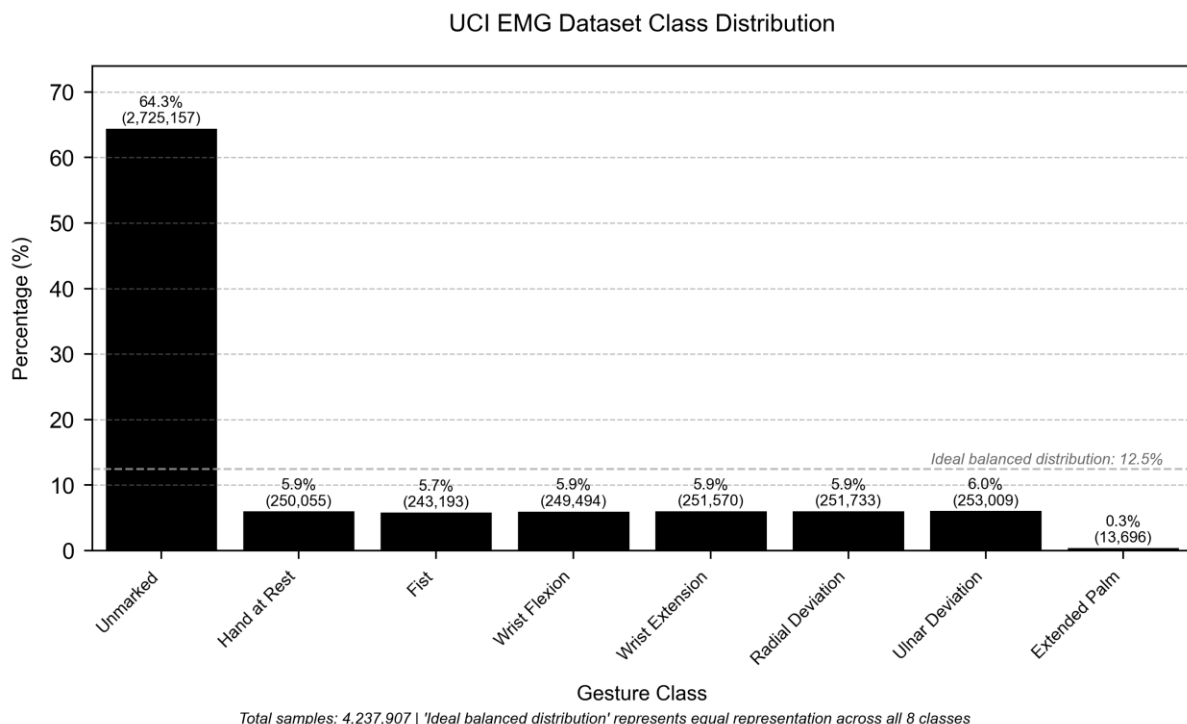


Figure 4 UCI EMG Dataset Class Distribution

Figure 4 illustrates the class distribution in the UCI EMG dataset. The most striking observation is the significant imbalance between different classes. The "Unmarked" class (class 0) represents 64.3% of the total samples, while the actual gesture classes (1-7) each account for approximately 5.7-6.0% of the data. The "Extended Palm" gesture (class 7) has notably fewer samples (0.3%), as not all participants performed this gesture.

This pronounced class imbalance has important implications for the development and evaluation of classification algorithms. The ideal balanced distribution, shown by the dashed line at 12.5%, represents what would be expected if all eight classes had equal representation. The significant deviation from this balanced distribution necessitates careful consideration in the model training process to prevent bias toward the majority class.

In the following chapters, the dataset is processed separately for each pipeline based on the differing requirements of gesture classification and user identification.

4. Gesture Classification Pipeline

4.1 Pipeline Overview

The hand gesture classification system follows a structured and modular pipeline designed to transform raw EMG signals into accurately recognised hand gestures. This pipeline builds upon established practices in biosignal processing, while integrating several tailored enhancements to suit the characteristics of the UCI EMG dataset.

The system is composed of four main stages:

- **Data Acquisition and Loading:**

In the first stage, EMG recordings from the UCI dataset are collected, organised, and validated. To ensure data consistency, user identifiers are preserved, and the temporal order of recordings is maintained. Invalid or missing entries are handled systematically, and gesture labels are cross-checked against a predefined set to ensure correctness. To improve efficiency during repeated experiments, the processed data is stored in an intermediate format for quick reuse.

- **Signal Preprocessing:**

Preprocessing involves selecting relevant EMG channels, converting signal data into a standardised numerical format, and identifying issues such as abnormally high signal values or flat-line segments that may indicate sensor malfunction. Missing data points are filled using linear interpolation. This stage intentionally avoids complex filtering techniques to maintain low computational cost, relying instead on robust feature extraction to capture key signal patterns.

- **Feature Extraction:**

The pre-processed EMG signals are segmented using a sliding window approach, where each segment is transformed into a feature vector. These vectors combine multiple time-domain features—including mean absolute value, root mean square, variance, and statistical measures such as kurtosis and skewness—as well as autoregressive coefficients that capture signal dynamics. The use of parallel processing and result caching ensures that feature extraction remains efficient even across large-scale experiments.

- **Classification:**

The final stage of the pipeline involves gesture classification using a neural network model with a configurable structure, including mechanisms such as regularisation and normalisation to improve generalisation. Alternative machine learning models are also explored. Given the significant class imbalance in the dataset, a hybrid data balancing technique is applied, combining synthetic sample generation for minority classes with filtering strategies to remove noisy or ambiguous samples from the majority class.

This modular architecture allows each stage of the pipeline to be evaluated and optimised independently while maintaining consistency across the system.

Parameters such as window size, channel configuration, feature selection, and model architecture are all configurable, supporting a wide range of experimental conditions. This design enables systematic comparisons between different approaches while ensuring reproducibility and flexibility in model development.

4.2 Signal Preprocessing

Signal preprocessing plays a crucial role in ensuring data quality prior to feature extraction. The implemented pipeline focuses on verifying data integrity, selecting relevant sensor channels, and handling missing values, while deliberately omitting traditional EMG filtering techniques.

4.2.1 Channel Selection for Complexity Control

The UCI EMG dataset provides recordings from eight sensors positioned around the forearm. To investigate the effect of sensor quantity on classification performance

and model complexity, the pipeline supports configurable selection of channel subsets. Specifically, models were trained and evaluated using different combinations of 4, 5, 6, 7, and all 8 channels.

This selective use of sensors allows for controlled experiments that balance classification accuracy against computational efficiency. Although this reduces the number of input features, it is not a form of dimensionality reduction in the algorithmic sense (e.g., PCA or LDA), as no mathematical transformation is applied.

The timestamp column, which marks the time of each sample, is excluded during preprocessing. Temporal information is later incorporated during the windowing step in feature extraction, which captures sequential dependencies in the data.

4.2.2 Data Type Standardisation

To ensure numerical consistency and efficient processing, all EMG channel data are converted to 32-bit floating-point format. This representation balances precision with computational efficiency for EMG signal magnitudes.

4.2.3 Signal Integrity Verification

Instead of applying traditional filtering techniques, the pipeline includes validation checks to identify problematic segments. Extreme Values Detection: Signals exceeding ± 2000 μV are flagged as physiologically implausible and may indicate sensor error or detachment. Flat Signal Detection: Segments where more than 10% of consecutive samples remain constant are flagged as potential disconnection or digitisation artifacts. These checks improve diagnostic transparency without modifying the raw signal.

4.2.4 Missing Value Handling

Missing samples are addressed using linear interpolation, which estimates short gaps in the signal based on neighbouring values. This method preserves continuity in the signal and maintains the physiological relevance of the data.

4.2.5 Gesture Label Mapping

Numerical class labels are mapped to descriptive gesture names to improve interpretability during analysis and visualisation. This mapping does not affect model training but supports clearer reporting of results.

4.2.6 Omission of Traditional Filtering

The pipeline deliberately omits standard EMG filtering techniques such as bandpass or notch filtering[29]. This is because the original dataset documentation indicates that basic filtering was applied during acquisition, also statistical feature extraction methods used later in the pipeline offer inherent robustness to noise. And most importantly, retaining raw signal characteristics may allow the model to learn subtle discriminative patterns that could otherwise be removed by aggressive filtering.

4.3 Feature Extraction

Feature extraction is a critical stage in the hand gesture classification pipeline, responsible for transforming pre-processed EMG signals into compact and discriminative representations. These representations capture key patterns of muscle activity that distinguish one gesture from another. The adopted approach combines traditional time-domain features with autoregressive modelling, enabling the characterisation of both amplitude-based and temporal aspects of the signal within each segment.

4.3.1 Sliding Window Segmentation

The continuous EMG signal is segmented using a sliding window approach, which allows for localised analysis of the muscle activity. This method divides the signal into overlapping segments (windows), where each window represents a short temporal slice of the signal used for feature extraction.

The segmentation is parameterised by two configurable values: Window size (segment dimension), which defines the length of each segment, with tested values of 256, 512, 1024, and 2048 samples, and step size (increment dimension), which controls the overlap between consecutive windows, set to 64, 128, 256, or 512 samples.

This method creates a trade-off between temporal resolution and feature stability. Smaller windows offer better responsiveness to gesture transitions, while larger windows provide more stable feature estimates at the cost of reduced temporal precision. The window parameters were varied during experimentation to evaluate their effect on model performance and complexity.

To assign a class label to each segment, the most frequently occurring gesture label within the window (the statistical mode) is used. This reduces ambiguity near gesture transitions and ensures that each feature vector is representative of a single gesture.

4.3.2 Time-Domain Features

Integrated EMG (IEMG):

IEMG is the sum of absolute values of the EMG signal withing a window, reflecting the overall muscle activity. This can be expressed as,

$$IEMG = \sum_{n=1}^N |x_n| \quad (1)$$

Where $x[t]$ = EMG signal in the t^{th} time frame, N = total number of time frame of EMG Signal.

Mean Absolute Value (MAV):

Average of the absolute values, providing a normalised measure of signal amplitude that is less sensitive to outliers. This can be expressed as,

$$MAV = \frac{1}{N} \sum_{n=1}^N |x_n| \quad (2)$$

Root Mean Square (RMS):

Square root of the mean of the squared signal values, this is recognised as a robust descriptor of muscle contraction levels[30]. This can be expressed as,

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N x_n^2} \quad (3)$$

Logarithmic RMS (Log-RMS):

A log-transformed version of the RMS, introduced to compress the dynamic range of the EMG signal. A small positive threshold is enforced to avoid logarithm of zero or negative values, improving numerical stability.

$$\text{Log} - RMS = \log(RMS + \epsilon) \quad (4)$$

Where ϵ is a small number to avoid $\log(0)$.

Variance (VAR):

Measures the variability of the signal amplitudes, indicating the complexity and dispersion during different muscular activities. This can be expressed as,

$$VAR = \frac{1}{N-1} \sum_{n=1}^N |x_{n-u}|^2 \quad (5)$$

Where u is the sample mean.

Kurtosis:

Kurtosis is a higher-order statistical moment that measures the peakiness of the EMG amplitude distribution. High kurtosis values suggest strong, transient muscle activation events, reflecting rapid changes in contraction intensity during different gestures[31]. This can be expressed as,

$$Kurtosis = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_n - u}{\sigma} \right)^4 - 3 \quad (6)$$

Where σ is the distribution's standard deviation of x .

Skewness:

Describes the asymmetry of the amplitude distribution, highlighting biases in signal deviation during certain hand gestures. This can be expressed as,

$$Skewness = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_n - u}{\sigma} \right)^3 \quad (7)$$

4.3.3 Autoregressive Modelling

In addition to time-domain features, an autoregressive (AR) model is fitted to each EMG window to capture the signal's temporal and spectral characteristics. Based on [1], a 6th-order AR model is used, resulting in six AR coefficients per channel. This can be expressed as,

$$A_n = \sum_{p=1}^P a_p x_{n-p} + \omega_n \quad (8)$$

Where a_p = AR coefficients, ω_n = white noise and $P = 6$ (AR model order)

AR modelling provides an efficient alternative to full spectral decomposition (e.g., via Fourier transforms), modelling each EMG signal as a linear combination of its previous values. Error handling is incorporated to ensure that feature extraction remains robust, even if AR modelling occasionally fails for unstable segments.

4.3.4 Feature Vector Composition

The final feature vector for each window concatenates all extracted features across all active EMG channels. For the default setting with 8 channels:

$$\text{Feature Vector Dimension} = 8 \times (7 \text{ time domain features} + 6 \text{ AR coefficients}) = 104$$

The dimensionality scales linearly with the number of selected channels, allowing flexibility for feature selection or channel reduction experiments.

To maintain a consistent input structure for machine learning models, the feature extractor ensures that feature vectors are uniformly sized, even in cases of extraction failure (e.g., via feature padding or exclusion).

4.4 Class Imbalance Handling

Class imbalance presents a significant challenge in EMG-based gesture classification, particularly with the UCI dataset where certain gestures are substantially underrepresented. This section details the multistage resampling strategy implemented to mitigate this imbalance while preserving the integrity of the feature space.

4.4.1 Characterisation of Class Imbalance

As shown in Figure 4, the UCI EMG dataset exhibits pronounced class imbalance, with the "unmarked" class (class 0) heavily overrepresented and certain gesture classes, particularly the "extended palm" (class 7), severely underrepresented. This imbalance biases classification algorithms towards the majority class, compromising recognition performance for minority gestures.

An initial analysis of the class distribution during a typical experimental run reveals that the unmarked segments (class 0) constitute approximately 64% of the dataset (5,311 out of 8,274 samples), while the extended palm gesture (class 7) represents only 0.34% of the samples (28 out of 8,274). This reflects an extreme imbalance ratio of roughly 190:1 between the most and least represented classes. Such severe imbalance creates a challenging learning environment, where models may achieve superficially high overall accuracy while performing poorly on rare gestures.

4.4.2 Data Balancing Strategy

To address the severe class imbalance in the UCI dataset, a three-stage data balancing strategy was developed and applied. The process included: (1) adaptive majority class undersampling, (2) adaptive minority class oversampling, and (3) synthetic sample cleaning. This approach aimed to create a more uniform class distribution while preserving the meaningful structure of the original data.

4.4.2.1 Adaptive Majority Class Undersampling

The first stage involved controlled undersampling of the majority class (class 0, corresponding to "unmarked" movements). The target sample count for the majority class was calculated based on the mean sample count of all meaningful gesture classes (classes 1–7). Specifically, the new target for the majority class was set to three times the minority class average to maintain representation without allowing it to dominate the dataset.

Formally, given the class counts, the target was calculated as:

$$Target_{maj} = \min(count_0, 3 \times \text{mean}(count_1, count_2, \dots, count_7)) \quad (9)$$

In the experimental setup using the UCI dataset, the majority class initially contained 5,311 samples, while the average minority class size was approximately 423 samples. Consequently, the target majority class size was computed as 1,269 samples.

A RandomUnderSampler[32] was applied with a fixed random seed to ensure reproducibility, using a strategy that under sampled only the majority class while retaining all minority classes unchanged. This controlled reduction significantly improved the balance between the classes without entirely discarding the majority class data.

4.4.2.2 Adaptive Minority Class Oversampling

Following majority class reduction, the second stage focused on boosting the representation of minority classes through oversampling. BorderlineSMOTE[33], an advanced variant of the Synthetic Minority Over-sampling Technique (SMOTE), was employed to generate synthetic samples specifically near class decision boundaries, where misclassification risk is highest.

A dynamic scaling approach was implemented, where the degree of oversampling was inversely proportional to the original class size: Classes with fewer than 50 samples received up to 6× oversampling, classes between 50 and 200 samples received approximately 3× oversampling, larger minority classes were conservatively increased by up to 2×.

The adaptive oversampling strategy was formalised by calculating a separate target for each minority class based on its initial size, ensuring realistic synthetic data generation.

In the experimental run: Class 7 ("Extended Palm"), with only 28 samples, was oversampled to 168 samples. Other classes were proportionally scaled according to their original counts.

To perform effective oversampling, BorderlineSMOTE requires two key parameters: 'k_neighbours', which controls the number of nearest neighbours used to generate synthetic samples, and 'kind', which determines the specific behaviour for selecting "borderline" samples for synthesis.

In this implementation, 'k_neighbors' was dynamically adapted based on the size of the smallest minority class to ensure that synthetic samples could still be generated even for extremely rare gestures. The final value was set to 5, balancing sufficient neighbour information while avoiding overfitting.

The 'kind' parameter was set to 'borderline-1', a conservative strategy where synthetic samples are created only for minority examples classified as "danger" those surrounded by a majority of other-class samples. This choice prioritised robustness and minimised the risk of introducing noisy synthetic points into the dataset.

4.4.2.3 Synthetic Sample Cleaning

The third and final stage involved cleaning the synthetic samples to remove potential noise and sharpen class decision boundaries. EditedNearestNeighbours (ENN) [34] was applied after oversampling, systematically removing any sample whose three nearest neighbours predominantly belonged to a different class.

The application of ENN served multiple purposes: It eliminated synthetic samples generated in ambiguous or overlapping regions, it helped sharpen the separation

between classes, and it reduced the likelihood of overfitting to artifacts introduced by oversampling.

Following the cleaning step, the "unmarked" class (class 0) was further reduced from 1,269 to 472 samples, and the smallest minority class (Extended Palm) stabilised at 157 samples.

4.4.3 Theoretical Foundation and Justification

The resampling approach in this work tackles the classic imbalanced learning challenge described by [35]: undersampling risks losing majority-class information, while naive oversampling often leads to overfitting.

The implemented strategy combines targeted techniques to address this problem. Controlled undersampling is applied first to reduce the overwhelming majority class while preserving its essential structure. Then, BorderlineSMOTE is used to create synthetic samples specifically near decision boundaries, rather than throughout the feature space. For extremely rare classes (like "extended palm"), a more aggressive oversampling is applied, where more conservative treatment is applied for better-represented classes. Finally, ENN cleaning removes ambiguous synthetic points to sharpen class boundaries.

This sequential approach creates a balanced training set that improves classification performance while maintaining realistic data representation.

4.4.4 Implementation Outcome

The final class distribution demonstrates a substantial improvement in balance, lowering the imbalance ratio from 190:1 to approximately 5.3:1.

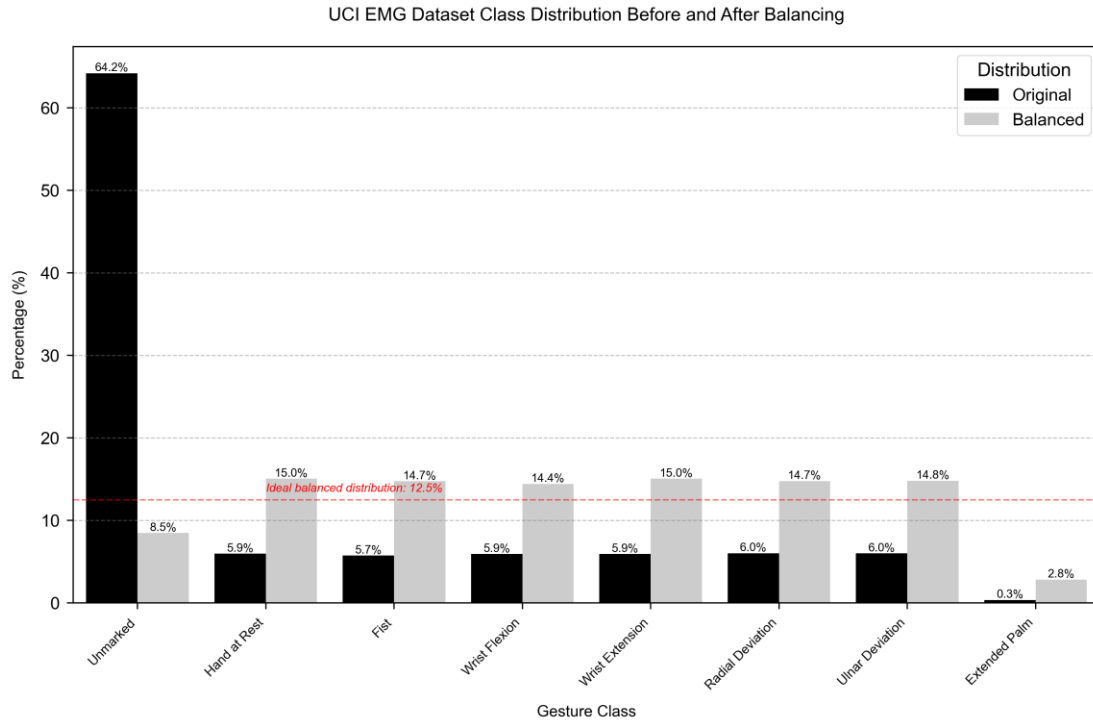


Figure 5 UCI EMG dataset class distribution before and after balancing

Figure 5 illustrates the full evolution of class distributions before and after applying the complete resampling pipeline.

4.5 Hyperparameter Tuning

Hyperparameter tuning is a critical step in optimising the performance of machine learning models. For this work, a systematic approach was employed to identify an optimal neural network architecture and training configuration. The objective was to maximise validation accuracy while maintaining model simplicity and preventing overfitting.

4.5.1 Tuning Strategy

The tuning process utilised the Hyperband algorithm [36], it was implemented through the Keras Tuner library [37]. Hyperband offers an efficient search mechanism by dynamically allocating computational resources to promising model configurations and early stopping underperforming trials. This enabled a broad exploration of potential neural network architectures.

The search space was designed to capture a range of model complexities and learning behaviours. The hyperparameters tuned included:

- Number of hidden layers: between 2 and 5 layers
- Units per layer: 8 to 128 neurons, varying by layer
- Dropout rates:
 - First hidden layer: between 0.0 and 0.5
 - Subsequent hidden layers: between 0.0 and 0.4
- Learning rate: logarithmically sampled between 1e-4 and 1e-2

4.5.2 Tuning Process

The tuning procedure evaluated 253 different model configurations, trained with default settings of using all 8 EMG channels with window size of 2048 and an increment of 512 samples. Each trial involved training a candidate network for up to 100 epochs, with early stopping triggered based on stagnation in validation accuracy to prevent unnecessary computation and overfitting.

Hyperband dynamically managed computational resources across multiple rounds ("brackets"), allocating more epochs to promising configurations while eliminating weaker models early. This adaptive resource allocation ensured a balance between exploration of new configurations and exploitation of high-performing models.

Validation accuracy served as the primary objective for evaluating each trial. All experiments were conducted using the same validation dataset to ensure consistency and comparability across trials.

4.5.3 Tuning Results

Upon completion of all trials, the hyperparameter search identified several high-performing models. The top-ranked configurations are summarised in Table 1. All top five models achieved validation accuracies exceeding 98.6%, demonstrating the robustness of the search process.

Table 1 Top 5 hyperparameter configurations ranked by validation accuracy

Rank	Validation Accuracy	Number of Layers	Learning Rate	Layers Architecture	Layers Dropout Rate
1	98.92%	4	0.00157	[128, 48, 64, 24]	[0.3, 0.3, 0.0, 0.4]
2	98.86%	4	0.00102	[128, 32, 48, 40]	[0.2, 0.0, 0.1, 0.3]

3	98.74%	3	0.00103	[128, 24, 48]	[0.4, 0.2, 0.0]
4	98.72%	2	0.00299	[72, 30]	[0.4, 0.0]
5	98.68%	2	0.00138	[128, 56]	[0.3, 0.0]

4.5.4 Model Selection

The hyperparameter search revealed that while the top-performing model with four hidden layers achieved a validation accuracy of 98.92%, a simpler two-layer architecture achieved a highly comparable 98.72% accuracy, differing by only 0.2%.

Given this minimal performance difference, the simple two-layer network was selected for further evaluation and deployment. This model configuration reduces computational complexity and training time while also decreasing the risk of overfitting which offers a more robust solution for practical applications.

4.6 Experimental Analysis and Results

This section presents a comprehensive evaluation of our EMG-based hand gesture classification pipeline, analysing how various parameter configurations affect classification performance.

4.6.1 Experimental Setup

The experimental framework systematically evaluated multiple configurations by varying three parameters: channel count and subset, window sizes, and two different models (scikit-learn and TensorFlow).

For channel configurations, five different EMG channel subsets were used (ranging from 4 to 8 channels), based on recommendations from [1]. From Section 4.3.1, four different sliding window sizes were evaluated (256, 512, 1024, and 2048 samples), with corresponding increment sizes of 64, 128, 256, and 512 samples, respectively.

Each unique combination resulted in 40 distinct experimental configurations (5 channel sets $\times$ 4 window sizes $\times$ 2 model types). All experiments used consistent preprocessing and feature extraction methods as described in previous sections.

For each experimental configuration, the following metrics were recorded: accuracy, f1-score, precision and recall. These metrics provide a comprehensive view of classification performance across different configurations.

4.6.2 Results

Impact of Channel Configuration

Figure 6 shows how the number of channels affects classification accuracy across different window sizes and model types.

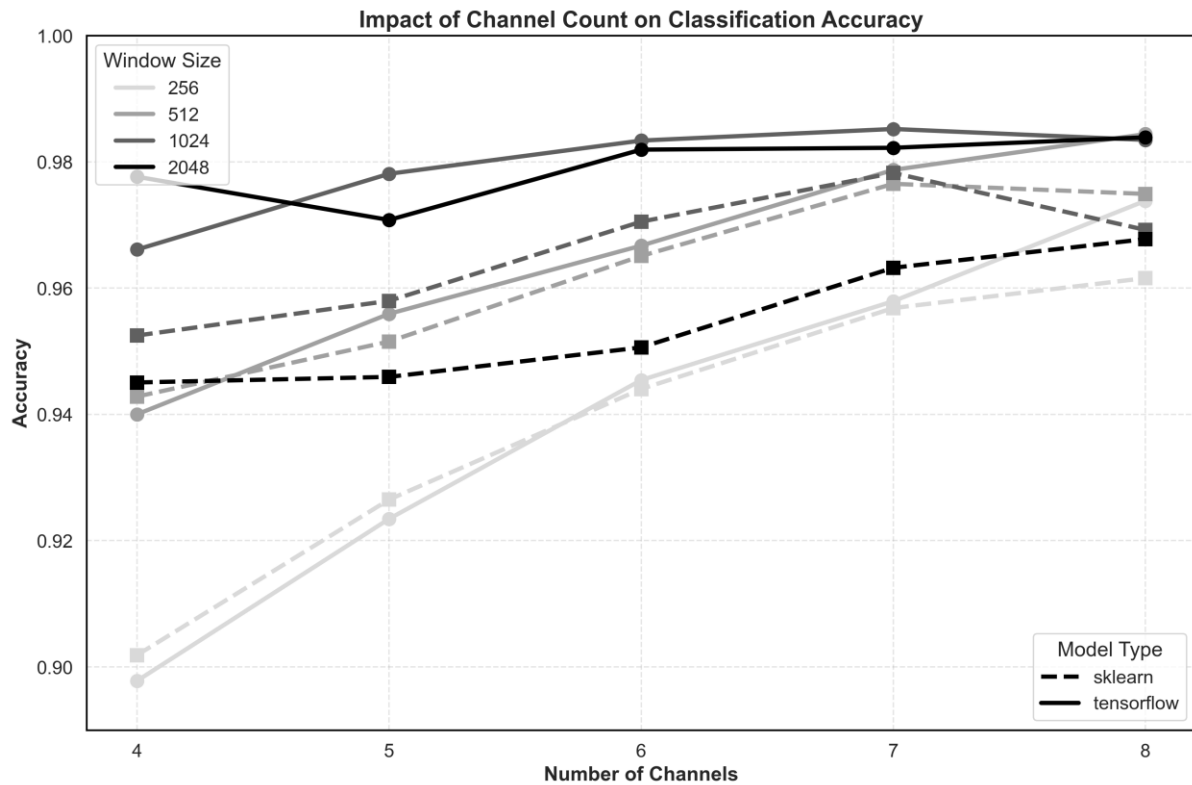


Figure 6 Impact of channel count on classification accuracy. Solid lines represent TensorFlow models, while dashed lines represent scikit-learn models. Different shades represent different window sizes.

As shown in Figure 6, accuracy consistently improves with additional channels regardless of window size or model architecture. The most significant improvements occur when increasing from 4 to 6 channels, with more modest gains when moving from 6 to 8 channels.

With 4 channels, accuracy ranges from 89.8% to 97.7%, depending on window size and model type. With 8 channels, accuracy ranges from 94.5% to 98.5%. The 7-channel configuration demonstrates performance nearly equivalent to the 8-channel setup, with accuracy differences of less than 0.7%

Effect of Window Size

Figure 7 shows how window size impacts classification accuracy across different channel configurations.

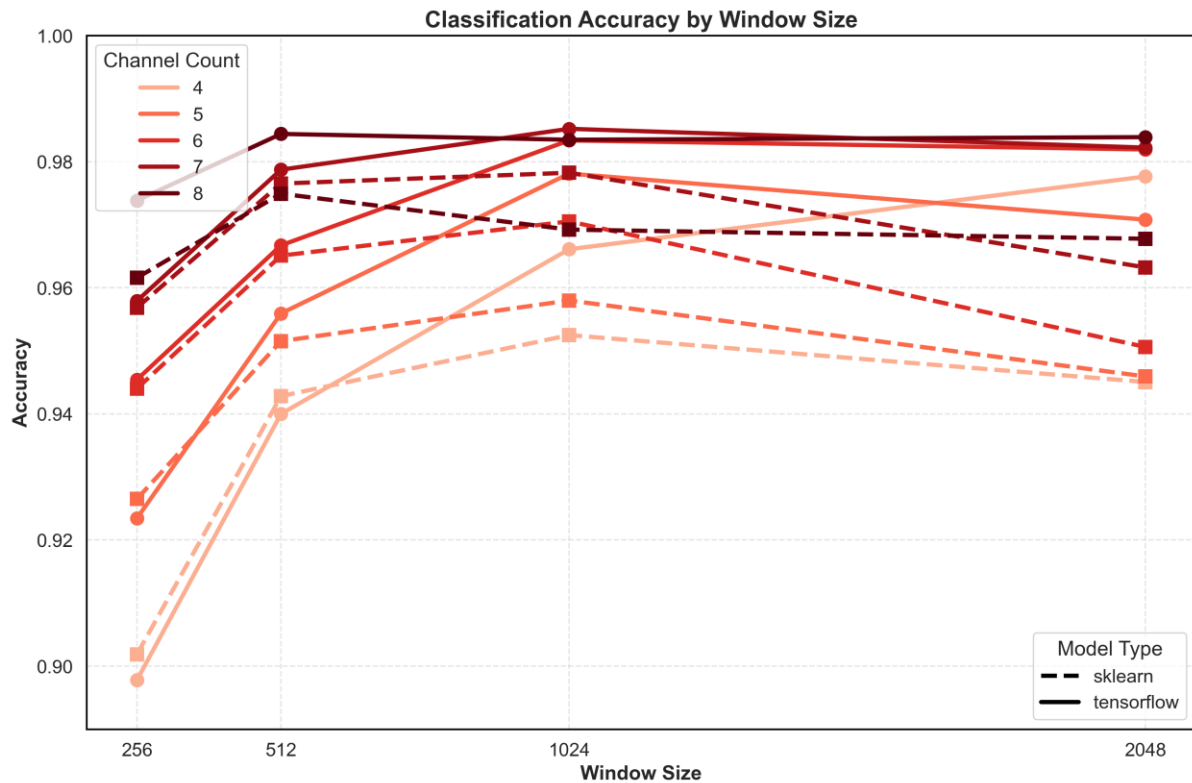


Figure 7 Classification accuracy by window size. Solid lines represent TensorFlow models, while dashed lines represent scikit-learn models. Different shades represent different channel counts.

Notably, larger window sizes consistently produce better classification results across all channel configurations. The improvement is most dramatic when increasing from 256 to 512 samples, with diminishing returns beyond 1024 samples.

The TensorFlow model shows greater resilience to small window sizes compared to the scikit-learn model. With 8 channels, even the smallest window size (256) achieves over 94% accuracy with the TensorFlow model

Model Comparison

A direct comparison was performed between the TensorFlow-based model and the scikit-learn model across various channel configurations and window sizes.

Table 2 summarises the classification accuracies obtained with each model and the difference in performance between them.

Table 2 Classification accuracy and F1 comparison between scikit-learn and TensorFlow models across different configurations.

Channels	Window Size	TF Accuracy	SKL Accuracy	Diff (TF-SKL)	F1 Diff	Better Model

4	2048	97.76%	94.50%	3.26%	3.13%	TensorFlow
4	1024	96.61%	95.25%	1.36%	1.80%	TensorFlow
4	256	89.78%	90.19%	-0.41%	-1.98%	scikit-learn
4	512	94.00%	94.28%	-0.28%	-0.24%	scikit-learn
5	2048	97.08%	94.59%	2.49%	2.57%	TensorFlow
5	1024	97.81%	95.80%	2.01%	2.55%	TensorFlow
5	512	95.59%	95.15%	0.44%	0.21%	TensorFlow
5	256	92.34%	92.65%	-0.31%	-1.84%	scikit-learn
6	2048	98.19%	95.06%	3.13%	3.33%	TensorFlow
6	1024	98.34%	97.05%	1.28%	1.83%	TensorFlow
6	512	96.67%	96.51%	0.16%	0.23%	TensorFlow
6	256	94.54%	94.40%	0.14%	-0.58%	TensorFlow
7	2048	98.22%	96.32%	1.90%	1.79%	TensorFlow
7	1024	98.52%	97.82%	0.70%	0.72%	TensorFlow
7	512	97.87%	97.65%	0.22%	0.30%	TensorFlow
7	256	95.79%	95.68%	0.11%	-0.91%	TensorFlow
8	2048	98.39%	96.78%	1.61%	1.66%	TensorFlow
8	1024	98.35%	96.92%	1.43%	1.74%	TensorFlow
8	256	97.38%	96.16%	1.23%	1.52%	TensorFlow
8	512	98.44%	97.49%	0.95%	1.17%	TensorFlow

TensorFlow models outperformed scikit-learn models in 17 out of 20 configurations. The average performance advantage of TensorFlow models was approximately +1.07%. The maximum observed advantage was +3.26% (in the 4-channel, 2048-sample window configuration). In three configurations, scikit-learn models slightly outperformed TensorFlow models, with the maximum observed sklearn advantage being -0.41%.

TensorFlow-based models consistently achieved higher classification accuracies, particularly in configurations involving larger window sizes and higher numbers of channels. The scikit-learn models showed good performance for smaller models and shorter windows but struggled to match the deep learning models in more complex settings.

The superior performance of TensorFlow models can be attributed to their ability to model non-linear feature interactions more effectively through multi-layer

architectures, whereas the scikit-learn models (based on classical machine learning methods) were comparatively limited in their representational capacity. This advantage became evident as the complexity of the input feature space increased, such as with larger numbers of EMG channels and extended temporal windows.

4.7 Complexity Overview

The complexity model from [1] provides a systemic way to quantify computational complexity by calculating the number of operations performed for the feature extraction process and the MLP training.

$$Features = u[P(2N + 21) + 17N + 24]$$

$$MLP = [q - 1][l(3r + 6)]$$

$$Total = u[P(2N + 21) + 17N + 24] + [q - 1][l(3r + 6)] \quad (10)$$

Table 3 Complexity parameters and values taken

Parameters	Definition	Taken Value
u	Number of channels	4,5,6,7,8
N	Width of window	256, 512, 1024, 2048
P	Order of AR model	6
q	Number of Classes	8
k	Width of time-domain feature vector	13
r	Number of neurons in MLP	110
l	Number of hidden layers in MLP	2

Complexity Calculation for Best Configurations

Table 4 Complexity calculation for best performing configuration for each EMG channel count

Channels (u)	Window (N)	Features Complexity	MLP Complexity	Total Operations	Accuracy
4	2048	221,624	73,040	294,664	97.76%
5	1024	138,515	73,040	211,555	97.81%
6	1024	166,218	73,040	239,258	98.34%

7	1024	193,921	73,040	266,961	98.52%
8	512	110,812	73,040	183,852	98.44%
8	2048	443,248	73,040	516,288	98.72%

The 8-channel model with 2048-sample window has the highest computational complexity and requires 516,288 operations per classification, this configuration achieves the highest observed accuracy. However, the 7-channel model with a 1024-sample window offers a strong trade-off, achieving 98.52% accuracy while reducing computational complexity by nearly half (266,961 operations).

This highlights the trade-off between model complexity and performance: increasing the number of channels and enlarging the window size improves classification accuracy but at the cost of greater computational demand. In real-time practical applications, such trade-offs must be considered carefully.

5. User Identification Pipeline

5.1 Pipeline Overview

The EMG-based user identification system is designed to transform raw EMG signals into unique biometric signatures for accurate user identification. This pipeline follows Section IV in [11]. It integrates tailored enhancements to suit the characteristics of the UCI EMG dataset and the specific requirements of biometric identification.

The system involves these steps:

- **Data Loading and Signal Preprocessing**

Since the same UCI EMG dataset is used, this follows the same logic in Section 4 for the gesture classification pipeline.

- **Segmentation**

EMG signals are divided into fixed-length windows using configurable parameters for window size and overlap. With the UCI dataset's 200 Hz sampling rate, each window spans 5 seconds (1000 samples), with a 1-second (200 samples) overlap between consecutive windows. This segmentation creates discrete analysis units that capture

complete muscle activation patterns while maintaining temporal relationships. The 5-second window provides sufficient temporal context to identify user-specific characteristics in muscle activation, while the 1-second overlap ensures smooth transitions between windows and prevents information loss at segment boundaries. For identification, consistent window sizes are critical to ensure comparable feature extraction across different recording sessions and subjects.

- **Feature Extraction**

The core of the identification pipeline lies in its feature extraction approach, which computes Root Sum Square (RSS) values for each EMG channel to characterise individual muscle activation patterns. Unlike gesture classification that relies on diverse time-domain feature sets, biometric identification leverages the inherent uniqueness in how each person's neuromuscular system activates during standard movements. These features form distinctive "EMG signatures" that remain consistent for an individual across different recording sessions.

- **Feature Projection**

To improve classification performance, the pipeline applies Kernel Fisher Discriminant (KFD) transformation[38], which maximises separation between different users while minimising variation within the same user's samples. This biometric-focused dimensionality reduction enhances the discriminative power of the feature space[39].

- **Biometric Identification**

The final stage employs a neural network classifier that maps EMG feature patterns to specific user identities. The model architecture is optimised for multi-class identification with cross-validation ensuring generalisation across different recording sessions. Performance evaluation focuses on user-specific metrics, analysing how consistently each subject can be recognised and which gestures provide the most reliable biometric information.

5.2 Feature Extraction and Projection

The biometric user identification system relies on distinguishing unique neuromuscular activation patterns among individuals. To achieve this, the pipeline

employs specialised feature extraction techniques followed by advanced projection methods to maximise discriminative power.

5.2.1 Feature Extraction with Root Sum Square

At the core of the feature extraction process is the Root Sum Square (RSS) calculation, RSS is used for capturing muscle contraction levels during different hand movements, giving a different scaling for each person. For each EMG channel, the RSS is computed as:

$$s_{RSS} = \sqrt{\sum_{n=1}^N |s[n]|^2} \quad (11)$$

where $s[n]$ represents the normalised EMG signal at sample point n . The UCI dataset used is already normalised, this provides an advantage in maintaining consistent feature scales across all subjects.

For each window segment, the RSS coefficients from all eight EMG channels are concatenated to form an 8-dimensional feature vector. This vector serves as a compact biometric signature for that specific segment of muscle activity.

5.2.2 Kernel Fisher Discriminant for Feature Projection

The Root Sum Square (RSS) features provide a foundation for user identification, there is a need for further enhancement to maximise the separability between different users. This pipeline employs Kernel Fisher Discriminant (KFD) analysis. KFD is a non-linear dimensionality reduction technique specifically suited for multi-class classification problems as introduced in [38].

KFD transforms the original feature space into a new subspace where samples from the same user are clustered closely together while samples from different users are pushed farther apart. Unlike linear projection methods that often struggle with complex multi class separation, KFD uses a kernel function (e.g., polynomial, radial basis function (RBF), or linear) to map features into a higher dimensional space where separation is achieved.

To prevent data leakage, KFD is integrated into the model training process. First, the KFD transformer is fitted exclusively on the training data, then the fitted transformation is applied separately to the validation and test data, and finally the trained KFD model is preserved for consistent application to new samples during

prediction. This approach maintains methodological rigor, which could otherwise result in overly optimistic performance estimates. By ensuring that information from the test set does not influence the training process, the evaluation metrics provide a realistic measure of the system's generalisation ability.

5.3 Biometric Identity Definition

A fundamental methodological question arose regarding the definition of a user's biometric identity. As mentioned in 3.1, the UCI dataset includes recordings from 36 subjects, each performing hand gestures with both their left and right arms in separate recording sessions. This presents two possible approaches to defining biometric identity:

- **Per-Arm Approach:** Treating each arm as a distinct biometric identity, resulting in up to 72 identities (two per subject).
- **Per-Subject Approach:** Treating each subject as a single identity, regardless of the arm used, resulting in 36 identities.

To determine the most appropriate approach, a comparative experiment was conducted using a representative subset of the dataset.

5.3.1 Comparative Analysis

Both approaches were implemented using identical feature extraction techniques, model architectures, and training protocols. The pipeline uses a scikit-learn Multi-Layer Perceptron (MLP) classifier, two layers with 150 neurons in the hidden layers, with Kernel Fisher Discriminant (KFD) transformation applied to the feature space.

The experiment involved 12 subjects, yielding 24 arm-based identities or 12 subject-based identities. Gestures that are labelled 1–7 were included in the evaluation excluding unmarked data (class 0). Performance was assessed using standard 80/20 train-test splits and 5-fold cross-validation.

Table 5 Overall accuracy comparison between per-arm and per-subject approach

Metric	Per-Arm	Per-Subject	Difference
Overall Test Accuracy	86.89%	94.09%	+7.21%
Cross-Validation Accuracy	82.59% $\pm$ 3.03%	92.14% $\pm$ 1.47%	+9.55%

The per-subject approach showed a marked improvement in both overall test accuracy (+7.21%) and cross-validation accuracy (+9.55%). The per-subject model had greater consistency, with a lower cross-validation standard deviation (1.47% vs. 3.03%).

Table 6 Gesture specific performance comparison between per-arm and per-subject

Gesture - Label	Per-Arm Accuracy	Per-Subject Accuracy	Difference
Hand at rest – 1	75.00% (36 samples)	74.29% (35 samples)	-0.71%
Fist – 2	97.14% (35 samples)	94.74% (38 samples)	-2.41%
Wrist flexion – 3	83.72% (43 samples)	100.00% (37 samples)	+16.28%
Wrist extension – 4	80.00% (45 samples)	93.48% (46 samples)	+13.48%
Radial deviations – 5	94.59% (37 samples)	100.00% (43 samples)	+5.41%
Ulnar deviations – 6	90.91% (44 samples)	100.00% (36 samples)	+9.09%
Extended palm – 7	100.00% (4 samples)	100.00% (2 samples)	0.00%

The per-subject approach achieved perfect classification accuracy (100%) for gestures 3, 5, and 6, and showed substantial improvements for gesture 4. Minor decreases were observed for gestures 1 and 2; however, these were outweighed by improvements in most gestures.

5.3.2 Discussion and Selection

Based on the experimental results, it was concluded that defining biometric identity on a per-subject basis is significantly more effective for EMG-based user identification. This is due to the higher overall and cross-validation accuracies, lower variance in cross-validation results, perfect classification on several gesture types, and a closer alignment with real-world biometric applications, where a user would be identified consistently regardless of which arm is being used.

The pipeline supports both approaches with the default being the per-subject approach. This approach was adopted for all subsequent experiments and analyses in this work.

5.4 Data Limitations and Augmentation

The UCI dataset presents challenges for biometric user identification, these challenges arise from intrinsic limitations in data distribution and sample availability. This section examines those constraints in more detail and describes the augmentation strategies developed to overcome them.

5.4.1 Sample Scarcity

As discussed in Chapter 3, the UCI dataset suffers from class imbalance, with around 63% of samples belonging to the “Unmarked” class (class 0). Since the user identification system is based on distinct wrist-hand gestures rather than random muscle activity, all unmarked samples were excluded from training. This exclusion resulted in a reduction in volume of usable data.

After filtering, the dataset contained only 240 gesture segments out of an initial 650 (across five users), a reduction of less than half. On average, this yielded around 48 segments per user, further distribution across seven different gesture classes. The limited number of samples per user increased the risk of model overfitting and affected the model's ability to capture natural inter-session variability.

5.4.2 Data Augmentation Strategy

A controlled data augmentation strategy was developed to mitigate the sample scarcity in the UCI dataset. The goal is to increase the number of training examples while preserving the physiological plausibility of EMG signals.

Class-Balanced Augmentation

The augmentation process first analysed the class distribution of the biometric IDs to determine how many synthetic samples should be generated for each user. It used the user class with the highest number of samples as a reference, and additional synthetic samples were generated for users with fewer samples to achieve relative class balance. A maximum augmentation limit was applied per user to prevent excessive duplication.

This class-aware augmentation ensured that minority classes were reinforced without overinflating the dataset disproportionately.

Signal Transformation Techniques

Each synthetic sample was generated by applying a series of transformations to the original EMG feature vectors. These modifications were progressively scaled to introduce increasing variation while maintaining signal realism.

First, Gaussian noise was added to simulate sensor noise and electrical interference, with noise levels ranging between 1% and 3% of the mean absolute signal amplitude.

Second, amplitude scaling was applied using factors between 95% and 99%, to emulate variations in muscle contraction intensity and electrode positioning.

Finally, random jitter, adding small random perturbations of $\pm 2\%$, this was applied to simulate natural variations in activation timing and muscle recruitment.

The problem of inadequate availability for EMG signals is a common problem, these techniques are inspired by [40], which explores data augmentation of surface EMG.

5.4.3 Augmentation Outcome

After augmentation, the size of the dataset approximately doubled, resulting in 453 samples compared to the original 240. The class distribution among users was significantly more balanced, with each biometric ID represented by approximately 50 to 100 samples.

This augmentation strategy resulted in a great improvement in model performance. Specifically, the identification accuracy (for five users, all gestures) increased from 70.59% (without augmentation) to 89.01% (with augmentation).

The number of augmentations applied could be configured during training to facilitate comparative experiments on the impact of augmentation.

5.5 Experiments and Results

This section presents a detailed evaluation of the EMG-based user identification pipeline. To assess performance across different levels of classification complexity, a series of experiments is conducted with varying the number of users included in identification.

5.5.1 Experimental Setup

The experiment employed a multi-run design to address both subject variability and the randomness in user selection. Structured around different user group sizes: 5, 10, 15 and 20 users. For each of these configurations, ten independent runs were done. Each run selected a random subset of users drawn from the UCI dataset which contains data for 36 users. A standard pipeline configuration was consistently applied, which included Root Sum Square (RSS) for feature extraction, Kernel Fisher Discriminant (KFD) transformation with 10 components, and the data augmentation techniques detailed in Section 5.4 were enabled. Across all runs, the collected performance metrics included the overall identification accuracy, and the accuracy achieved for each individual gesture.

5.5.2 Overall Accuracy vs. Number of Users

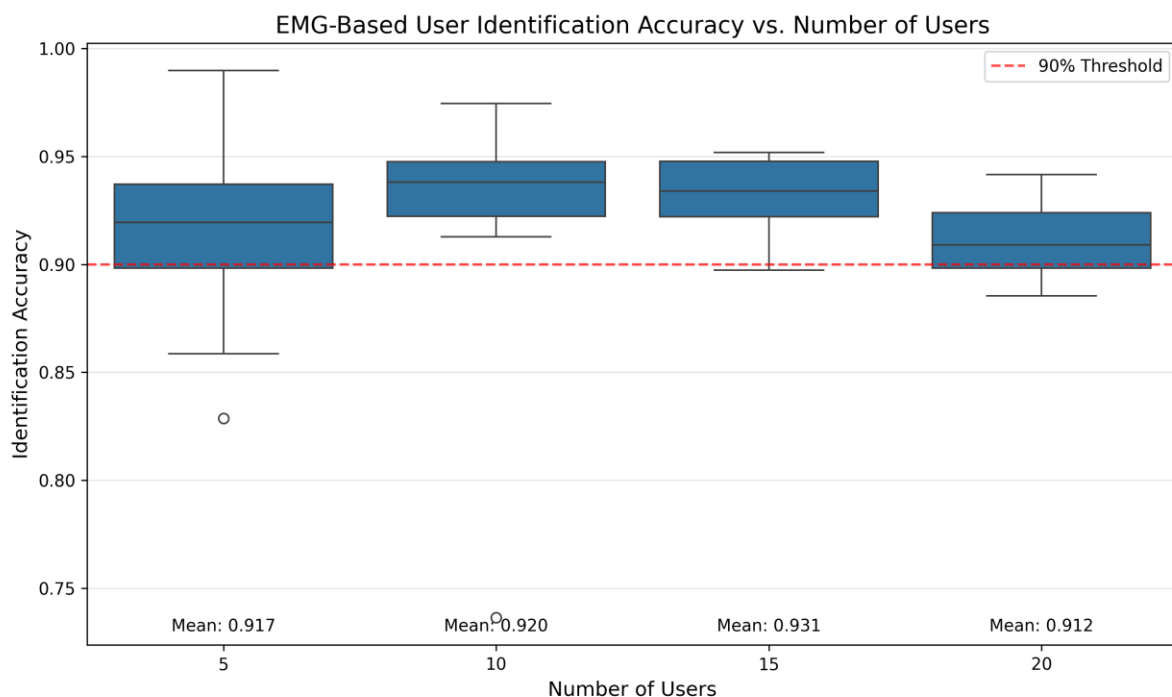


Figure 8 User identification accuracy vs. number of users

Figure 8 presents a box plot showing the distribution of identification accuracies across 10 runs for each user group size.

Each box represents 10 runs with randomly selected users. The red dashed line indicates a 90% accuracy threshold.

The results indicate a strong identification performance across all configurations. The mean accuracy remained above 91% for all user group sizes.

The 5-user configuration has the highest variance and one visible outlier, likely due to certain combinations being harder to distinguish based on EMG features.

The 10-user and 15-user configurations showed the most consistent performance, with narrow interquartile ranges and all runs exceeding the 90% threshold. For the 20-user configuration, a slight decrease in mean accuracy (91.2%) and increased variability were observed, which aligns with expectations as classification complexity increases with more classes.

The system shows robust scalability, it maintains high accuracy with the increase of number of users.

5.5.3 Gesture-Specific Identification Accuracy

Each gesture accuracy was examined across different user group sizes. The bar chart in Figure 9 shows the average classification accuracy for each gesture, with the number of users increasing from 5 to 20.

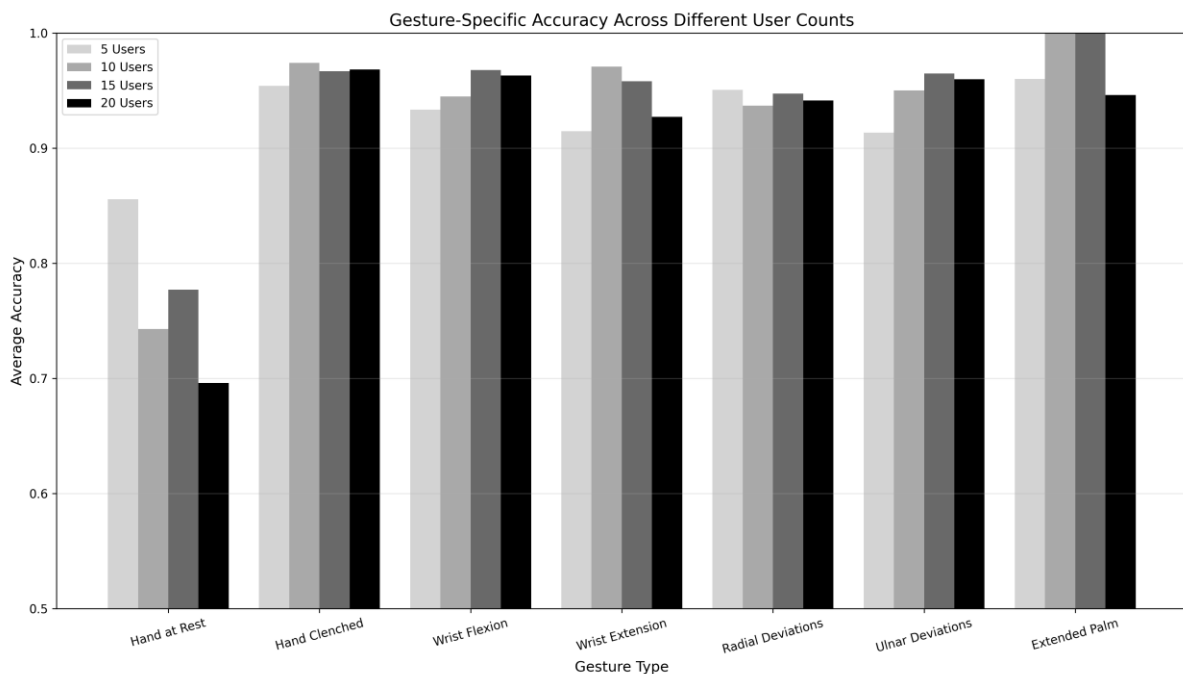


Figure 9 *Gesture specific accuracy across different user counts*

The "Hand at Rest" gesture demonstrated the poorest biometric performance, with accuracy declining from 85.6% with 5 users to just 69.6% with 20 users. This

degradation stems from the minimal myoelectric activity present during resting states, providing insufficient discriminative features for robust user separation in larger populations.

In contrast, dynamic gestures involving distinct muscle recruitment patterns maintained high identification accuracy. "Hand Clenched" exhibited exceptional consistency (94-97% accuracy) across all user group sizes, while "Wrist Flexion" and "Ulnar Deviations" similarly maintained strong performance above 93% even with 20 users. The "Extended Palm" gesture achieved perfect classification (100%) in the 10-user and 15-user groups, though performance decreased to 94.6% with 20 users. This suggests this gesture produces highly individual EMG signatures that remain separable until reaching larger population sizes where some pattern overlap may emerge.

5.6 Refining Model Performance through Gesture Selection

The user identification pipeline was refined by excluding the Hand at Rest gesture from both the training and testing datasets. From Figure 9, gesture-specific analysis indicated that this gesture consistently underperformed across multiple users, potentially limiting the model's overall classification accuracy. Given that the Hand at Rest state may not offer strong user-distinguishing features due to its low muscular activation, it was excluded to improve identification performance.

To maintain consistency and enable fair comparison, the same experimental protocol was followed as in earlier evaluations: the pipeline was executed for four user group sizes (5, 10, 15, and 20 users), with 10 randomised runs per group. Identification accuracy was recorded for each run.

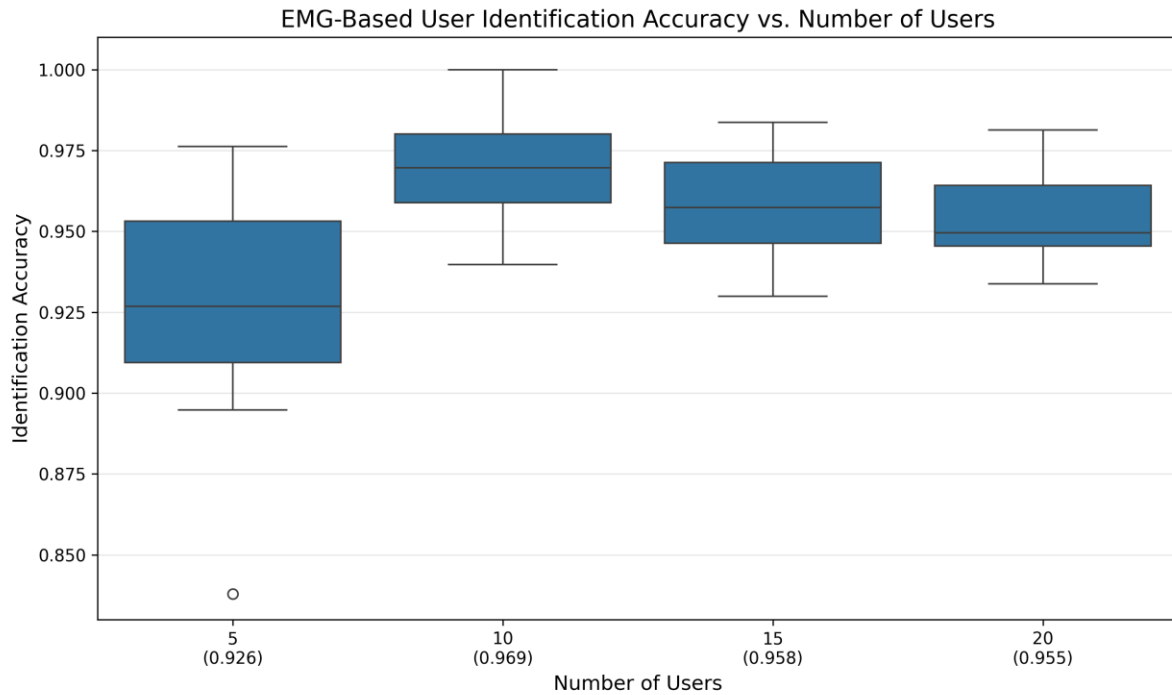


Figure 10 User identification accuracy vs. number of users excluding Hand at Rest gesture

Figure 10 shows the results; it shows the distribution of identification accuracy across the different user group sizes. The mean accuracy for each group is indicated in parentheses.

Notable improvements can be noticed compared to the previous configuration (which included all gestures). The spread of accuracy values across runs is tighter, this suggests improved model stability and reduced sensitivity to user selection when poor-quality gestures are excluded.

Filtering out less discriminative gestures can have a measurable positive impact on system performance. Gesture selection is a valuable consideration in the design of EMG-based biometric systems, especially in constrained-use scenarios where only a subset of gestures is available or practical.

The radar chart in Figure 11 illustrates the classification accuracy of individual hand gestures, evaluated across varying user pool sizes (5, 10, 15, and 20 users). The Hand at Rest gesture was excluded from this analysis.

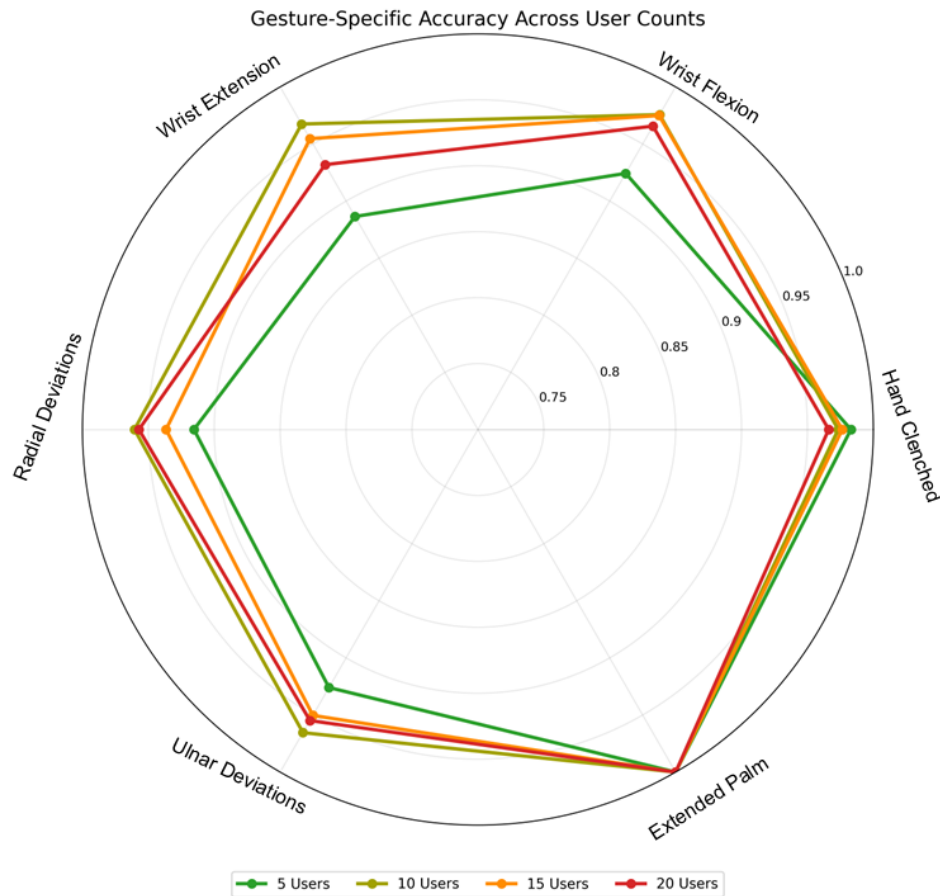


Figure 11 Gesture specific accuracy across user counts

The Hand Clenched and Extended Palm gestures consistently achieve the highest identification accuracy across all user group sizes, maintaining performance above 95% even with 20 users.

Radial and Ulnar Deviations show a positive response when scaling the number of users, this suggests that some gestures are more scalable for large-population identification systems.

Despite variation in performance across individual gestures, the overall accuracy remains high for most movements, even at the largest tested population size. This reinforces the robustness of the EMG-based user identification approach and supports its potential for deployment in practical biometric systems.

5.6.1 Best Performing Model

The EMG-based user identification pipeline achieved its highest performance with a configuration that excluded Hand at Rest gesture (Gesture 1) and focused on a cohort of 15 users. This configuration yielded the highest accuracy.

Using scikit-learn MLP classifier with two-layer architecture containing 100 and 50 neurons respectively. It achieved an overall classification accuracy of 96.46%, with cross-validation accuracy of 96.45% ($\pm 1.00\%$), and individual fold scores ranging from 95.65% to 98.02%, reflecting excellent generalisation across subjects.

Gesture analysis revealed that certain hand movements produced more distinctive and reliable EMG signatures:

Gesture 6 (Ulnar Deviation) and Gesture 7 (Extended Palm) both achieved 100% classification accuracy, though it is noted that the latter had fewer samples.

Gesture 5 (Radial Deviation) also performed strongly with 98.21% accuracy.

Gesture 2 (Hand Clenched) achieved 96.55%, demonstrating high consistency.

Gestures 3 and 4 (Wrist Flexion and Extension) followed with 94.34% and 92.86% accuracy, respectively.

The model demonstrated strong computational efficiency, completing training in 17.65 seconds, underscoring its practicality for real-world biometric systems that demand both performance and speed.

6. Conclusion and Future Work

6.1 Summary of Contributions

This work explored the viability of electromyography (EMG) signals as a means for human-computer interaction (HCI), focusing on both hand gesture classification and biometric user identification. Two complete machine learning pipelines were developed to this end.

The first pipeline involved a complete reconstruction of a legacy gesture classification system. The outdated codebase was restructured using modern software engineering practices and enhanced with improved signal preprocessing techniques, robust class balancing strategies (e.g., BorderlineSMOTE and ENN), and a modular pipeline design. This improved gesture classification accuracy on the UCI EMG dataset from its previously unrepresentative performance of 95% to approximately 99%, while also offering configurable trade-offs between classification

complexity and performance via selective EMG channel usage and varying window sizes.

The second major contribution was the design and implementation of a user identification pipeline, a novel implementation inspired by methodologies proposed in existing literature[11], [26], [27]. Time-domain EMG features were extracted and projected using Kernel Fisher Discriminant Analysis (KFDA) to enable user-specific classification. The final model, an MLP classifier implemented in scikit-learn, achieved a peak accuracy of 96.46% across 15 users, outperforming the 91% benchmark reported in[11]. Gesture-specific evaluations revealed that biometric separability varied significantly with movement type, this highlights the importance of gesture selection in optimising performance.

Together, these two pipelines demonstrate that EMG signals can effectively support both gesture-based control and biometric identification in constrained hardware settings, opening opportunities for practical, real-world applications in wearable systems, prosthetics, and personalised interfaces.

6.2 Limitations

Despite the promising results, certain limitations must be acknowledged:

Dataset constraints, The UCI EMG dataset[6], while widely used, suffers from class imbalance and contains few gesture samples per user. This limited the ability to generalise model robustness across more diverse or realistic settings.

All evaluations were performed offline, with no integration into a real-time acquisition or deployment framework. As such, the system's responsiveness, latency, and complexity under real-world conditions remain unverified.

6.3 Future Work

Building on the findings of this work, future work could enhance the identification system through more thorough hyperparameter tuning for the MLP classifier. Real-time implementation and validation would provide insights into practical aspects like latency and power usage.

Integrating gesture classification and user identification into a unified system, this can unlock secure gesture-driven interfaces.

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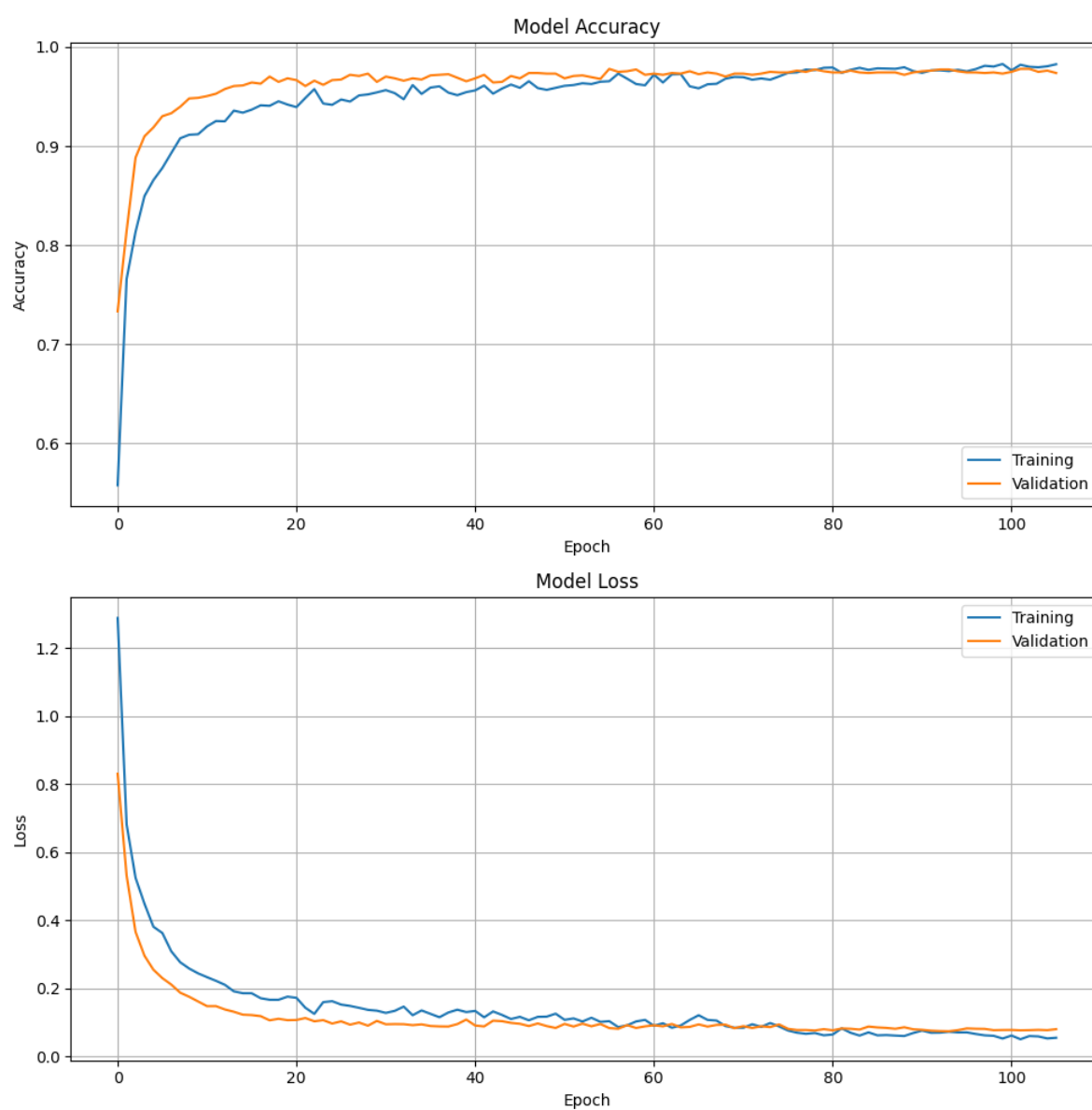
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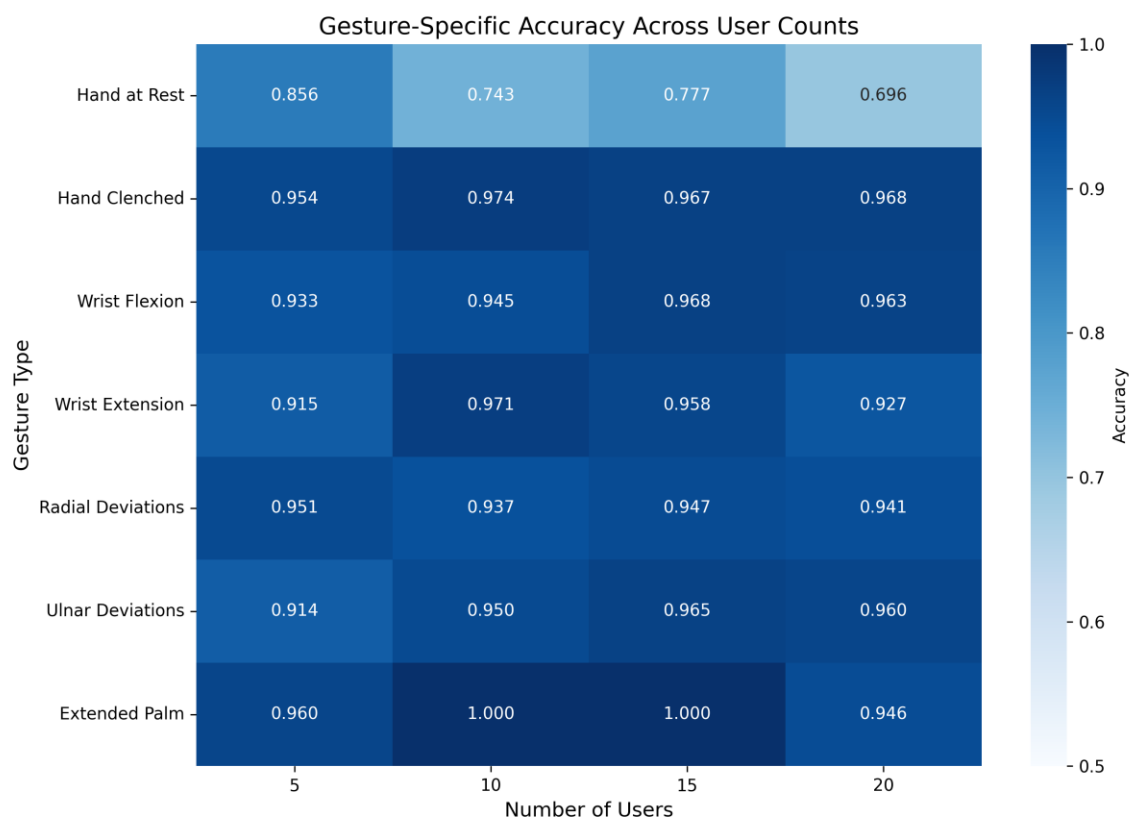
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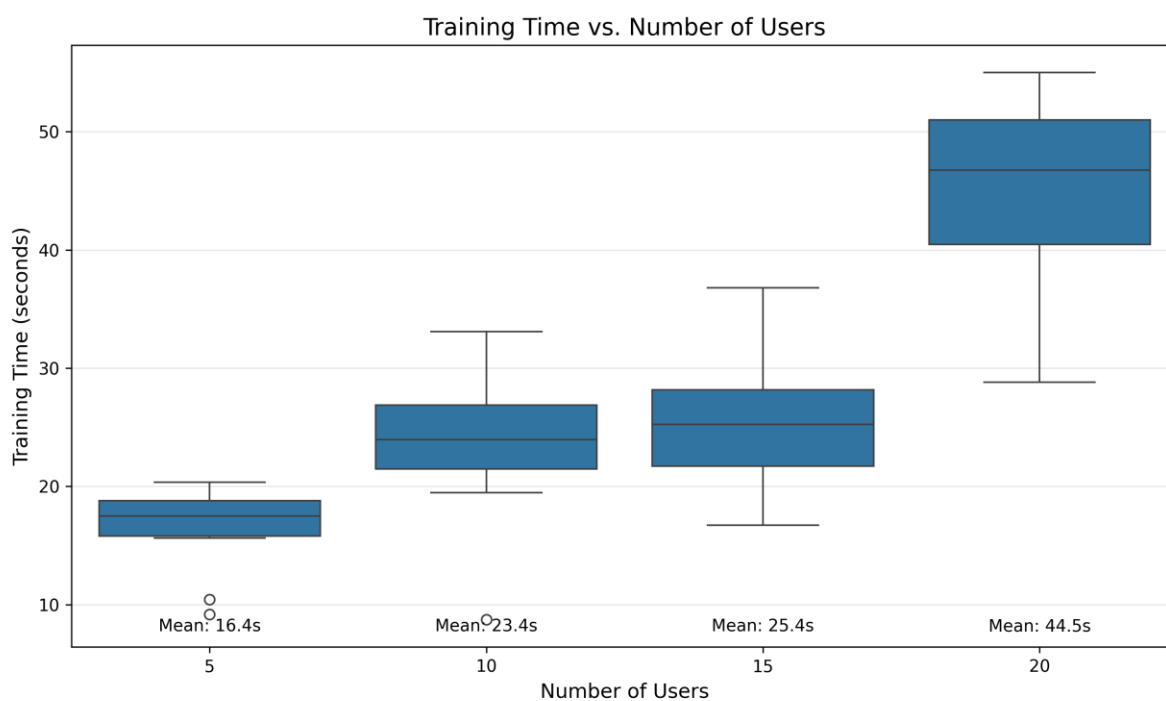
Appendices



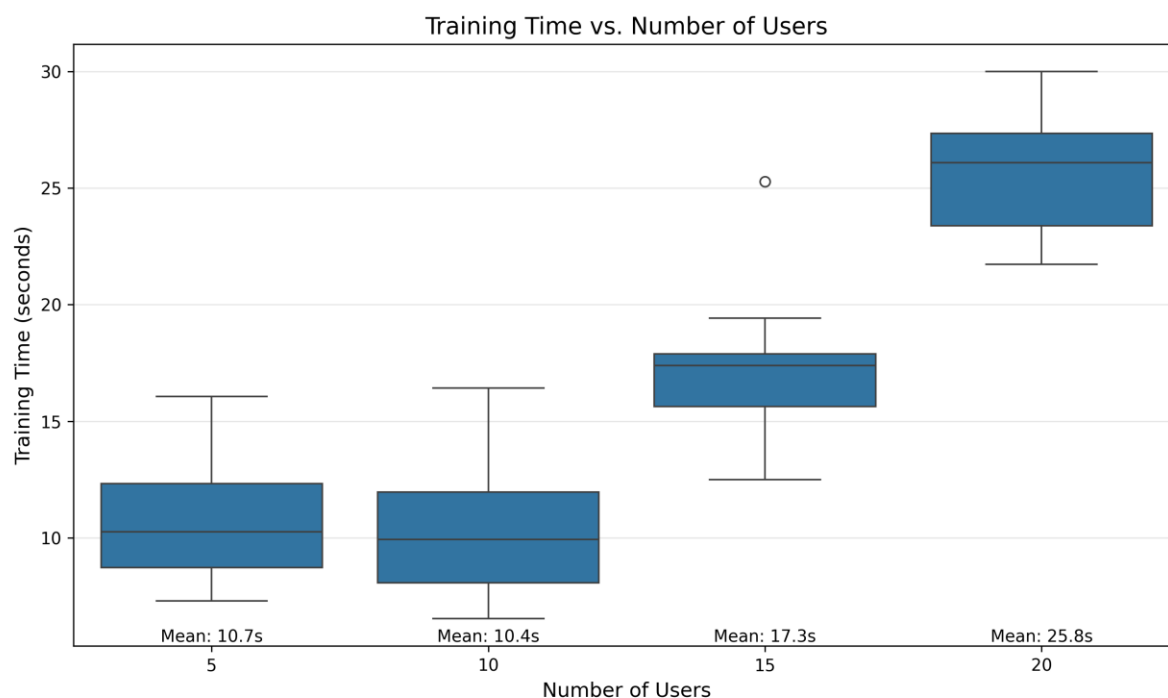
Appendix 1 Gesture classification model training history



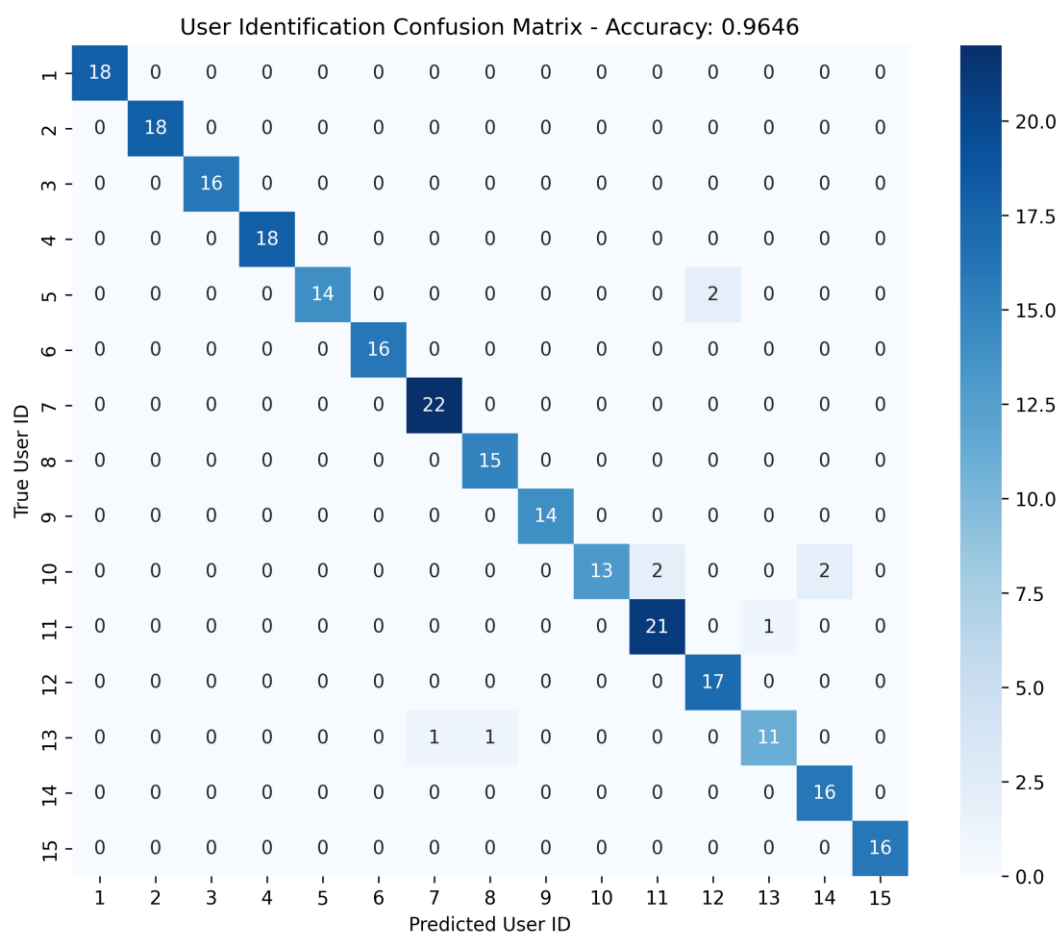
Appendix 2 User identification gesture specific accuracy across different number of users heatmap



Appendix 3 User identification pipeline training time for different number of users including all gestures



Appendix 4 User identification pipeline training time for different number of users excluding hand at rest gesture



Appendix 5 Confusion matrix for user identification model in 5.6.1